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CHARUSAT
BCA Semester-V**

CAUE302 : Python Programming

Practice Questions

Assignment-1

“Instructions: Write and execute the following programs in Jupyter Notebook (.ipynb).”

1. Write a Python program to store your name, city, and course in variables and display them.
 2. Create variables of type int, float, string, and bool and display their values and datatypes.
 3. Write a Python program to demonstrate dynamic typing by assigning an integer value to a variable and then changing it to a string.
 4. Write a Python program to accept your name and birth year, calculate your age based on the current year, and print your name with the calculated age.
 5. Write a Python program to assign the same value to multiple variables and display them.
 6. Write a Python program to accept length and breadth of a rectangle and calculate its area.
 7. Write a Python program to accept the radius of a circle and calculate its area.
 8. Write a Python program to perform all arithmetic operations (+, -, *, /, %, //, **) on two user-entered numbers.
 9. Write a Python program to calculate Simple Interest.
 10. Write a Python program to accept a student's Name, Course, and Semester and display all details in a formatted manner.
 11. Write a Python program to determine whether a student is eligible for examination based on the following conditions:
Attendance >= 75
AND
Internal Marks >= 20
Display the final result (True or False).
 12. Write a Python program to swap two numbers without using a third variable.
 13. Write a Python program to check whether a given character exists in a string using the Membership Operator.
 14. Write a Python program to calculate the square and cube of a number using the exponentiation operator.
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15. Accept a number from the user and classify it as Positive, Negative, or Zero.
 16. Accept a number from the user. If it is divisible by 5, print 'Divisible by 5'.
 17. Accept three numbers from the user and print the largest among them.
 18. Accept Name, Phone No, age and monthly salary from the user. Using nested if, determine loan eligibility:
Age >= 18 AND Salary >= 25000 :Loan Approved

Age ≥ 18 AND Salary < 25000 : Loan Rejected: Low Salary

Age < 18 : Not Eligible Age Below 18

19. Calculate Profit or Loss by accepting Cost Price and Selling Price from the user.

20. Accept a number N from the user. Using a while loop, print all numbers from 1 to N.

21. Accept a number N. Using a while loop, calculate and print the sum of all numbers from 1 to N.

22. Accept a number from the user and print its multiplication table from 1 to 10 using a for loop.

23. Accept a number N. Using a for loop, calculate and print the sum of all even numbers from 1 to N.

24. Accept a number N from the user. Using a for loop, calculate and print the factorial of N.

25. Accept the number of terms N. Using a while loop, print the Fibonacci series.

Enter number of terms: 8	0 1 1 2 3 5 8 13
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26. Accept a number from the user. Using a for loop, check if the number is Prime or Not Prime.

27. Accept a year from the user. Check whether it is a Leap Year or a Non-Leap Year.

28. Accept a number from the user. find and print all the factors of that number.

Ex: **Sample Input:**

Plaintext

Enter number: 12

Expected Output:

Plaintext

Factors of 12 are: 1 2 3 4 6 12